

- Large cap stocks finished the day down 3.2% from the prior night's close and down 2.7% from the opening. This is equivalent to a 1.8 (close-to-close) and 1.5 (open-to-close) sigma event and is not something considered out of the norm in itself. The sigma-event measure is a standard z-score measurement and is computed by dividing the price movement by the daily volatility as a means to normalize across stocks. Z-scores or sigma events between ± 1 are considered normal movement. Z-scores between ± 2 , while high, are still considered relatively normal since 95% of the observations should be between a z-score of ± 2 . Theoretically, z-scores should only be > 3 or < -3 0.27% of the time.
- Small cap stocks finished the day down 3.8% from the prior night's close and down 3.4% from the opening. This is equivalent to a 1.4 (close-to-close) and 1.2 (open-to-close) sigma event. Again, this is not something unexpected or something that should be considered out of the ordinary.
- Analysis of the high-low range, however, showed that the day was much out of the ordinary. The high-low percentage range measured as the change from the high of the day to the low of the day was -10.7% for large cap stocks and -11.8% for small cap stocks. This is equivalent to a z-score of 6.1 for large caps and 4.4 for small caps. To put this into perspective, we should only see price movement of this magnitude 1 in 7.5 million years for large cap and 1 in 738 years for small cap.
- Even though prices returns have much fatter tails than the normal distribution, this analysis shows that the intraday price movement was much different than ordinary. And then couple this price movement with an almost equal and opposite reversal and we can start to see the significance of this flash crash event.

The next phase of the flash crash analysis focused on evaluating the intraday behavior for the SPY exchange traded fund (ETF), as a proxy for the SP500 index and a sample of ten stocks. The analysis really shed light onto what happened on flash crash day.

The SPY volume on flash crash day was 3.3 times its 30 day average. This was much higher than the average for all stocks which was about 2 times its average. The close-to-close price change was -3.4% which was consistent with other stocks, but due to much lower volatility this movement was equivalent to a 3.5 sigma event ($Z = -3.5$). The high-low price change for SPY was -10.8% which was also consistent with the average stock high-low range (-10.7% for large cap stocks).

Figure 2.8 illustrates the intraday prices and volumes for SPY. Price remained relatively flat throughout the day. Then beginning at